

Course Code: B20HS4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B.Tech I Semester MODEL QUESTION PAPER					
UNIVERSAL HUMAN VALUES-2: UNDERSTANDING HARMONY					
(Common to AIDS, AIML, CSBS, CSE, CSG, IT & ME)					
Time: 3 Hrs.			Max. Marks:70M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
		UNIT-I			
1	a).	Discuss natural acceptance	1	2	7
	b).	Differentiate prosperity and deprivation	1	2	7
		OR			
2	a).	Write a note on physical facilities	1	2	7
	b).	Deliberate the right understanding in perspective to self-exploration.	1	2	7
		UNIT-II			
3	a).	Illustrate coexistence of "I" and "Body".	2	2	7
	b).	Explain doer, seer and enjoyer.	2	2	7
		OR			
4	a).	Discuss Characteristic activities of Harmony with "I".	2	2	7
	b).	Explain Sanyam and Health.	2	2	7
		UNIT-III			
5	a).	Write a note on human-human relationship as regarding harmony.	3	2	7
	b).	Differentiate intention and competence.	3	2	7
		OR			
6	a).	Discuss salient values in relationship.	3	2	7
	b).	Illustrate universal Harmonious Society - an Undivided society.	3	2	7
		UNIT-IV			
7		Discuss orders of life in nature and its significance self-regulation of individual	4	2	14
		OR			
8		Illustrate existence of human being as coexistence with universe in perspective of space	4	2	14
		UNIT-V			
9		Discuss importance of professional competence for augmenting universal	5	3	14

		human order.			
		OR			
10	a).	Case study of typical holistic technologies.	5	3	7
	b).	Role of engineer in promoting harmony in society	5	3	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks



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Course Code: B20AM4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B. Tech. I Semester MODEL QUESTION PAPER					
ROBOTIC PROCESS AUTOMATION					
(Common to AIML and CSG)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
		UNIT-I			
1.	a).	Explain about RPA and compare its benefits with traditional automation techniques.	1	2	7
	b).	Describe how the benefits of RPA can be utilized in a business case study.	1	2	7
		OR			
2.	a).	Outline the key components of an RPA system.	1	2	7
	b).	Compare RPA with traditional automation methods.	1	2	7
		UNIT-II			
3.		Explain the different types of activities available in RPA tools like UiPath.	2	2	14
		OR			
4.	a).	Develop an RPA workflow using different control flow statements to handle decision-making.	2	3	7
	b).	Develop a flowchart showing how the components of an RPA system interact.	2	3	7
		UNIT-III			
5.	a).	Explain the concept of data scraping in RPA.	3	2	7
	b).	Explain how selectors are used to identify UI elements in RPA.	3	2	7
		OR			
6.	a).	Compare image-based automation and keyboard-based automation in RPA.	3	2	7
	b).	Utilize anchors to automate data extraction from a PDF document.	3	3	7
		UNIT-IV			
7.		Illustrate different types of triggers available in RPA for handling user events.	4	2	14

		OR			
8.	a).	Summarize common strategies for troubleshooting issues in RPA.	4	2	7
	b).	Explain the workflow that incorporates exception handling mechanisms to manage errors effectively.	4	2	7
		UNIT-V			
9.		Describe how servers are used to manage and deploy bots in an RPA environment.	5	2	14
		OR			
10.	a).	Describe the steps to publish a workflow in UiPath.	5	2	7
	b).	Demonstrate a strategy for managing and deploying updates in an RPA project.	5	3	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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Course Code: B20CS4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B.Tech. I Semester MODEL QUESTION PAPER					
CLOUD COMPUTING					
(Common to CSE, AIML and CSG)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
		UNIT-I			
1.	a).	List out & explain the different types of cloud computing service models	1	2	7
	b).	Explain Cloud essential characteristics	1	2	7
		OR			
2.	a).	Describe cloud computing architecture	1	2	7
	b).	Explain cloud deployment models	1	2	7
		UNIT-II			
3.	a).	Illustrate the relationship between virtualization and cloud computing	2	2	7
	b).	Interpret how to implement levels of virtualization	2	2	7
		OR			
4.	a).	What is a hypervisor? Explain XEN hypervisor architecture.	2	2	7
	b).	Illustrate how virtualization used in Data-Center Automation.	2	2	7
		UNIT-III			
5.	a).	Explain coordination of specialized autonomic performance managers	3	3	7
	b).	Determine the relationship between the scale of a system and the policies and the mechanisms for resource management. In your arguments, consider also the geographic scale of the system.	3	3	7
		OR			
6.	a).	Interpret the concept of Resource Bundling	3	2	7
	b).	Determine the limitations and potential challenges of implementing the Start Time Fair Queuing algorithm at scale in a large cloud-computing environment. What strategies employed to mitigate these challenges?	3	3	7
		UNIT-IV			
7.	a).	Summarize the storage models in cloud computing.	4	2	7
	b).	Explain the rationale behind secure cloud software requirements for a	4	2	7

		cloud-based application that handles sensitive customer data.			
		OR			
8.	a).	Explain the primary objectives of cloud information security, and why are they crucial in cloud computing?	4	2	7
	b).	Explain Google file system with a diagram	4	2	7
		UNIT-V			
9.	a).	Illustrate Map-Reduce design in Hadoop	5	2	7
	b).	Explain primary advantages of using Google App Engine for application deployment compared to traditional server provisional methods	5	2	7
		OR			
10.	a).	Explain the concept of the four levels of federation in cloud computing	5	2	7
	b).	Interpret the future of cloud federation	5	2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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Course Code: B20CD4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B.Tech. I Semester MODEL QUESTION PAPER					
NATURE INSPIRED COMPUTING TECHNIQUES					
COMPUTER SCIENCE AND DESIGN					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
UNIT-I					
1.	a).	Discuss Simulated Annealing Algorithm	1	2	7
	b).	Discuss in detail about Hyper optimization	1	2	7
OR					
2.	a).	Explain Ant and Bee Algorithms	1	2	7
	b).	Discuss about Stochastic Tunneling	1	2	7
UNIT-II					
3.	a).	Explain the Role of Genetic Operators in detail.	2	2	7
	b).	Explain in detail about Swarm Intelligence.	2	2	7
OR					
4.	a).	Discuss PSO Algorithm	2	2	7
	b).	Discuss variants in Differential Evolution	2	2	7
UNIT-III					
5.	a).	Discuss Standard Firefly Algorithm	3	2	7
	b).	Discuss about Variants of Cuckoo Search	3	2	7
OR					
6.	a).	Discuss Variations of Light Intensity and Attractiveness in Firefly Algorithm	3	2	7
	b).	Discuss about Cuckoo Breeding Behavior	3	2	7
UNIT-IV					
7.	a).	Discuss about Movement of Virtual Bats	4	2	7
	b).	Discuss the Variants of the Bat Algorithm	4	2	7
OR					
8.	a).	Discuss Loudness and Pulse Emission in Bat Algorithms.	4	2	7
	b).	Discuss any two applications of Bat Algorithms.	4	2	7
UNIT-V					
9.	a).	Discuss the various Characteristics of Flower Pollination	5	2	7

	b).	Discuss Single-Objective Design Benchmarks in the applications of Flower Pollination Algorithms	5	2	7
		OR			
10.	a).	Discuss Validation and Numerical Experiments on Flower Pollination Algorithms	5	2	7
	b).	Discuss Multi-Objective Design Benchmarks in the applications of Flower Pollination Algorithms	5	2	7
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks



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Course Code: B20AM4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B.Tech. I Semester MODEL QUESTION PAPER					
NoSQL DATABASES					
(Common to AIML and CSG)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
		UNIT-I			
1.	a).	Explain aggregates and the consequences of aggregate orientation	1	2	7
	b).	Explain impedance mismatch and how it is been handled	1	2	7
		OR			
2.	a).	Write a brief note on aggregate oriented databases	1	2	7
	b).	Explain the emergence of NoSQL	1	2	7
		UNIT-II			
3.	a).	Write a short note on types of Distributed models	2	2	7
	b).	What is consistency? Explain the various forms of consistency.	2	2	7
		OR			
4.	a).	Write a short note on Replication methods	2	2	7
	b).	Explain CAP theorem with an example	2	2	7
		UNIT-III			
5.	a).	Explain the features of Key-Value Store	3	2	7
	b).	Determine why shopping cart data uses key-value database as the best fit	3	3	7
		OR			
6.	a).	Determine how key-value store is suitable for storing session information	3	3	7
	b).	Explain when not to use Key-Value stores	3	2	7
		UNIT-IV			
7.	a).	Explain the features of document databases	4	2	7
	b).	Explain suitable uses cases for column-family data stores	4	2	7
		OR			
8.	a).	Explain suitable uses cases for document databases	4	2	7

	b).	Explain the features of column-family data store	4	2	7
		UNIT-V			
9.	a).	Explain the features of graph databases	5	2	7
	b).	Explain schema changes in RDBMS	5	2	7
		OR			
10.	a).	Briefly explain the use cases of graph databases	5	2	7
	b).	Explain schema changes in a NoSQL database	5	2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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Course Code: B20AM4105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B.Tech. I Semester MODEL QUESTION PAPER					
REINFORCEMENT LEARNING					
Common to AIML, CSG					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
UNIT-I					
1.	a).	Explain Limitations and Scope of Reinforcement Learning	1	2	7
	b).	Solve An Extended Example of Tic-Tac-Toe Problem	1	3	7
OR					
2.	a).	Explain the role of incremental updates in action-value methods for a k-armed bandit problem.	1	2	7
	b).	Apply optimistic initial values and UCB action selection in the 10-armed testbed and analyze their impact on exploration and exploitation.	1	3	7
UNIT-II					
3.	a).	Explain Agent-Environment Interface	2	2	7
	b).	Apply the concept of optimal policies to design a strategy for an agent in a continuing task within a Finite Markov Decision Process and evaluate its effectiveness.	2	3	7
OR					
4.	a).	Explain the process of policy evaluation and policy iteration in dynamic programming.	2	2	7
	b).	Implement value iteration for a given Markov Decision Process and analyze the efficiency of asynchronous dynamic programming	2	3	7
UNIT-III					
5.	a).	Explain the difference between Monte Carlo prediction and Monte Carlo estimation of action values.	3	2	7
	b).	Apply off-policy Monte Carlo control to optimize the policy of an agent and evaluate its performance compared to on-policy methods	3	3	7
OR					
6.	a).	What are the advantages of Temporal Difference (TD) prediction methods compared to Monte Carlo methods in reinforcement learning?	3	2	7
	b).	Apply Expected Sarsa to mitigate maximization bias in reinforcement learning algorithms and assess its impact on learning stability and	3	3	7

		efficiency.			
		UNIT-IV			
7.	a).	Illustrate why the Bellman Error is not Learnable	4	2	7
	b).	Apply the n-step Tree Backup algorithm to optimize the control of an agent in a complex decision-making scenario	4	3	7
		OR			
8.	a).	Compare and contrast the $sarsa(\lambda)$, Watkin's $Q(\lambda)$	4	2	7
	b).	Apply $TD(\lambda)$ to optimize an agent's policy in a complex decision-making scenario, examining the influence of λ values on learning speed and policy stability	4	3	7
		UNIT-V			
9.	a).	Illustrate The Policy Gradient Theorem.	5	2	7
	b).	Describe a personalized recommendation system using reinforcement learning and evaluate its performance compared to traditional methods.	5	2	7
		OR			
10.	a).	Discuss TD-Gammon and its Applications	5	2	7
	b).	Explain Actor-Critic Methods, Policy Gradient for Continuing Problem	5	2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B20CD4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B.Tech. I Semester MODEL QUESTION PAPER					
SOFTWARE PROJECT MANAGEMENT					
COMPUTER SCIENCE AND DESIGN					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
		UNIT-I			
1.	a).	Explain in detail various activities of software management	1	2	7
	b).	Explain few problems associated with software projects	1	2	7
		OR			
2.	a).	Explain in detail the five basic parameters of the Software Cost Model?	1	2	7
	b).	Explain detail about the effort estimation models.	1	2	7
		UNIT-II			
3.	a).	Define Artifact? Write short notes on Engineering Artifacts.	2	2	7
	b).	Illustrate the principles of modern software management	2	3	7
		OR			
4.	a).	Explain the typical minor milestones in the lifecycle of iteration	2	2	7
	b).	. Explain the first two phases of the life-cycle process.	2	2	7
		UNIT-III			
5.	a).	Illustrate in detail different models of COCOMO with an example	3	3	7
	b).	Illustrate periodic status assessments	3	3	7
		OR			
6.	a).	Write a short notes on software process workflow & iteration workflow	3	2	7
	b).	Sketch and explain in detail various steps in Project Monitoring Cycle	3	3	7
		UNIT-IV			
7.	a).	Discuss Cost monitoring with suitable example.	4	2	7
	b).	List and explain seven categories of resources	4	2	7
		OR			
8.	a).	Briefly explain about the earned values	4	2	7
	b).	Explain the risk assessment	4	2	7

		UNIT-V			
9.	a).	What is capability maturity model? Explain.	5	2	7
	b).	With neat diagram explain product and process metrics.	5	2	7
		OR			
10.		What is Software Quality Metrics? Explain in detail various Categories of Software Quality Metrics with suitable example each	5	2	14
CO-COURSE OUTCOME		KL-KNOWLEDGE LEVEL	M-MARKS		

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks



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IV B.Tech. I Semester MODEL QUESTION PAPER

BLOCKCHAIN TECHNOLOGIES

Common to AIML, CSG

Time: 3 Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

Assume suitable data if necessary

			CO	KL	M
		UNIT-I			
1	a).	Explain the benefits and limitations of blockchain technology.	1	2	6
	b).	Describe the various stages in Block chain evolution.	1	2	8
2	a).	Explain the type of players in the in Block Chain Ecosystem	1	2	8
	b).	Compare centralized applications with Decentralized applications.	1	2	6
		UNIT-II			
3	a).	Explain the Merkle-tree structure in Block chain.	2	2	6
	b).	Explain the Consensus algorithm in detail	2	2	8
		OR			
4	a).	Discuss about the types of Block chain nodes and the risks associated with block chain solutions.	2	2	6
	b).	Explain in detail the Life cycle of block chain transaction,	2	2	8
		UNIT-III			
5	a).	Illustrate the Block chain Relevance Evaluation Framework.	3	3	7
	b).	Describe the various types of Types of Block chain Applications	3	2	7
		OR			
6	a).	Determine the Architecture with Block chain Platforms.	3	3	8
	b).	Explain the approach for designing Blockchain applications.	3	2	6
		UNIT-IV			
7	a).	Explain the Ethereum Blockchain implementation.	4	2	8
	b).	Explain the Truffle framework and Ganache in Blockchain.	4	2	6
		OR			
8	a).	Describe smart contract programming.	4	2	7
	b).	Explain Tuna Fish Tracking Use Case with the help of Block chain	4	2	7

		Implementation.			
		UNIT-V			
9	a).	Explain the Hyper ledger Fabric Transaction Flow in detail.	5	2	7
	b).	Compare Hyper ledger with Ethereum	5	2	7
		OR			
10		Explain the Block chain implementation for Hyperledger	5	2	14
		CO-COURSE OUTCOME	KL-KNOWLEDGE LEVEL	M-MARKS	

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks



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Course Code: B20CD4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER VISION					
COMPUTER SCIENCE AND DESIGN					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
		UNIT-I			
1.	a).	Discuss in brief about Geometric Primitives and Transformation	1	2	7
	b).	Explain the terms Point Operators and Linear Filtering	1	2	7
		OR			
2.	a).	Illustrate Geometric Transformations and Global Optimization.	1	2	7
	b).	Explain Fourier Transforms in detail.	1	2	7
		UNIT-II			
3.	a).	Demonstrate Active Contours and Normalized Cuts	2	2	7
	b).	Explain 2D and 3D Feature based Alignment	2	2	7
		OR			
4.	a).	Describe in detail Mean Shift and Mode Finding with an example	2	2	7
	b).	Explain the terms Pose Estimation, Geometric Intrinsic Calibration	2	2	7
		UNIT-III			
5.	a).	Define Spline-based Motion and discuss in detail.	3	2	7
	b).	Discuss in detail about Two-frame Structure from Motion	3	2	7
		OR			
6.	a).	Demonstrate Layered motion in brief.	3	2	7
	b).	Describe Bundle Adjustment	3	2	7
		UNIT-IV			
7.	a).	Define Image Stitching and Explain image Matting and Compositing	4	2	7
	b).	Describe in detail about Global Alignment, Photometric Calibration	4	2	7
		OR			
8.	a).	Discuss about Texture Analysis and Synthesis	4	2	7
	b).	Explain about High Dynamic Range Imaging	4	2	7

		UNIT-V			
9.	a).	Illustrate Point-based Representation	5	3	7
	b).	Explain visualization along logarithmic axis.	5	2	7
		OR			
10.	a).	Discuss about Model-based Reconstruction	5	2	7
	b).	Explain Recovering Texture Maps with an example.	5	3	7
CO-COURSE OUTCOME		KL-KNOWLEDGE LEVEL	M-MARKS		

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Course Code: B20AM4110					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B.Tech. I Semester MODEL QUESTION PAPER					
SOCIAL NETWORK ANALYSIS					
Common to AIML, CSG					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
		UNIT-I			
1.	a).	Demonstrate Three Levels of Social Network Analysis	1	2	7
	b).	Describe Graph Visualization Tools	1	2	7
		OR			
2.	a).	Demonstrate Node Centrality	1	2	7
	b).	Compare Transitivity and Reciprocity	1	2	7
		UNIT-II			
3.	a).	Demonstrate Random network model	2	2	7
	b).	Compare Page Rank and Personalized Page Rank	2	2	7
		OR			
4.	a).	Demonstrate Watts-Strogatz Model	2	2	7
	b).	Compare SimRank and PathSim	2	2	7
		UNIT-III			
5.	a).	Explain Disjoint Community Detection	3	3	7
	b).	Identify Heuristic Models	3	3	7
		OR			
6.	a).	Explain Overlapping Community Detection	3	3	7
	b).	Identify Probabilistic Models	3	3	7
		UNIT-IV			
7.	a).	Demonstrate Epidemic Models	4	2	7
	b).	Describe Anomaly Detection in Static Networks	4	2	7
		OR			
8.	a).	Demonstrate Challenges in Anomaly Detection in Static Networks	4	2	7
	b).	Describe Independent Cascade Models	4	2	7

		UNIT-V			
9.	a).	Explain Machine Learning Pipelines	5	3	7
	b).	Demonstrate Malicious Activities on OSNs	5	2	7
		OR			
10.	a).	Explain GRL Pipelines	5	3	7
	b).	Demonstrate Modeling the Spread of COVID-19	5	2	7
CO-COURSE OUTCOME		KL-KNOWLEDGE LEVEL	M-MARKS		

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks



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Course Code: B20AM4111																																																								
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20																																																			
IV B. Tech. I Semester MODEL QUESTION PAPER																																																								
RECOMMENDER SYSTEMS																																																								
Common to AIML, CSG																																																								
Time: 3 Hrs.			Max. Marks: 70 M																																																					
Answer ONE Question from EACH UNIT																																																								
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Assume suitable data if necessary																																																								
			CO	KL	M																																																			
		UNIT-I																																																						
1.	a).	Explain the primary goals of a recommender system. How do these goals contribute to enhancing user experience in online platforms?	1	2	8																																																			
	b).	Describe any four real world applications of recommender systems	1	2	6																																																			
		OR																																																						
2.	a).	Describe the basic models of recommender systems.	1	2	7																																																			
	b).	Illustrate three domain-specific challenges faced by recommender systems.	1	2	7																																																			
		UNIT-II																																																						
3.	a).	Consider the following ratings table between five users and six items <table border="1"><thead><tr><th colspan="2"></th><th colspan="6">Item Id</th></tr><tr><th rowspan="6">User Id</th><th></th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr></thead><tbody><tr><td>1</td><td>5</td><td>6</td><td>7</td><td>4</td><td>3</td><td>?</td></tr><tr><td>2</td><td>4</td><td>?</td><td>3</td><td>?</td><td>5</td><td>4</td></tr><tr><td>3</td><td>?</td><td>3</td><td>4</td><td>1</td><td>1</td><td>?</td></tr><tr><td>4</td><td>7</td><td>4</td><td>3</td><td>6</td><td>?</td><td>4</td></tr><tr><td>5</td><td>1</td><td>?</td><td>3</td><td>2</td><td>2</td><td>5</td></tr></tbody></table> a) Compute the values of unspecified ratings of user 2 using user-based collaborative filtering algorithms. Use the Pearson correlation with mean-centering. Assume a peer group of size 2.			Item Id						User Id		1	2	3	4	5	6	1	5	6	7	4	3	?	2	4	?	3	?	5	4	3	?	3	4	1	1	?	4	7	4	3	6	?	4	5	1	?	3	2	2	5	2	3	7
		Item Id																																																						
User Id		1	2	3	4	5	6																																																	
	1	5	6	7	4	3	?																																																	
	2	4	?	3	?	5	4																																																	
	3	?	3	4	1	1	?																																																	
	4	7	4	3	6	?	4																																																	
	5	1	?	3	2	2	5																																																	
	b).	Given a high-dimensional user-item matrix, describe the steps you would take to apply SVD for reducing dimensionality and improving the performance of a neighborhood-based recommender system.	2	2	7																																																			
		OR																																																						
4.	a).	Demonstrate the first three iterations of personalized page rank computation to find the most similar users to target user 'u1' in the	2	3	7																																																			

		following rating matrix																																						
		<table><tr><td></td><td>I1</td><td>I2</td><td>I3</td><td>I4</td></tr><tr><td>U1</td><td>1</td><td></td><td></td><td>5</td></tr><tr><td>U2</td><td></td><td>5</td><td></td><td></td></tr><tr><td>U3</td><td>5</td><td>3</td><td></td><td>1</td></tr><tr><td>U4</td><td></td><td></td><td>3</td><td></td></tr><tr><td>U5</td><td></td><td></td><td></td><td>3</td></tr><tr><td>U6</td><td>5</td><td></td><td>4</td><td></td></tr></table>		I1	I2	I3	I4	U1	1			5	U2		5			U3	5	3		1	U4			3		U5				3	U6	5		4				
	I1	I2	I3	I4																																				
U1	1			5																																				
U2		5																																						
U3	5	3		1																																				
U4			3																																					
U5				3																																				
U6	5		4																																					
	b).	Explain the role of horting and predictability in predicting missing ratings in graph-based models for neighborhood-based methods	2	2	7																																			
		UNIT-III																																						
5.	a).	Demonstrate how Naïve Bayes Model can be used for Collaborative filtering with the help of an example	3	3	7																																			
	b).	Explain the process involved for using Decision and Regression trees for collaborative filtering	3	2	7																																			
		OR																																						
6.	a).	Demonstrate Non-negative matrix factorization with the help of an example	3	3	7																																			
	b).	Compare Neighborhood based collaborative filtering approach with Model based collaborative filtering approach.	3	2	7																																			
		UNIT-IV																																						
7.	a).	Illustrate constraint-based recommender systems	4	2	7																																			
	b).	Compare content based and collaborative filtering approaches for building recommender systems	4	2	7																																			
		OR																																						
8.	a).	Describe the basic components of a content-based recommender system.	4	2	6																																			
	b).	Summarize the methods used for feature representation, feature cleaning and feature selection in Content Based Recommender Systems	4	2	8																																			
		UNIT-V																																						
9.	a).	Explain the different Recommender Systems evaluation paradigms	5	2	7																																			
	b).	Summarize the general goals in evaluating the recommender systems.	5	2	7																																			
		OR																																						
10.	a).	Describe the metrics used for offline evaluation of recommender systems	5	2	7																																			
	b).	Explain the evaluation metrics that can be used in the case of implicit and explicit feedbacks.	5	2	7																																			

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B20AM4112					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B. Tech. I Semester MODEL QUESTION PAPER					
AI CHATBOTS					
Common to AIML and CSG					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
		UNIT-I			
1.	a).	Classify the elements of a customer-centric approach in financial services.	1	2	7
	b).	Illustrate the benefits of chatbots for a business.	1	2	7
		OR			
2.	a).	Describe case study to show the role of chatbots in the insurance industry.	1	2	7
	b).	Explain potential challenges companies might face in ensuring customer rights under the General Data Protection Regulation (GDPR).	1	2	7
		UNIT-II			
3.	a).	Apply chatbot-related terms to describe the main functions and roles in industries.	2	3	7
	b).	Select an AI-based approach and a Rule-based approach to solve a specific problem and compare their outcomes.	2	3	7
		OR			
4.	a).	Construct a real-world use case for chatbots, illustrating their application in a specific industry.	2	3	7
	b).	Identify the characteristics of customer service-centric chatbots and illustrate their benefits in improving customer support.	2	3	7
		UNIT-III			
5.	a).	Compare the functionalities and user experiences of chatbots versus traditional apps.	3	2	7
	b).	Explain the generic solution architecture for a private chatbot.	2	2	7
		OR			
6.	a).	Summarize the key success metrics used to evaluate chatbots.	2	2	7
	b).	Outline the benefits of chatbots in business operations.	3	2	7

		UNIT-IV			
7.	a).	Explain the architecture of chatbots.	4	2	7
	b).	Compare NLP, NLU and NLG in the context of chatbot technology.	4	2	7
		OR			
8.		Describe open-source NLP and NLU Tools for a real-world scenario.	4	2	14
		UNIT-V			
9.	a).	Choose the best tool among RASA, Microsoft Bot, and Google Dialog Flow for any use case and justify your choice.	5	3	7
	b).	Develop a simple QnA bot using QnA Maker and demonstrate its functionality.	5	3	7
		OR			
10.	a).	Illustrate LUCIS intent classification.	5	3	7
	b).	Illustrate the working of the RASA framework by building a simple chatbot and demonstrating its features.	5	2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks



Course Code: B20CD4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
IV B.Tech. I Semester MODEL QUESTION PAPER					
DATA VISUALIZATION					
COMPUTER SCIENCE AND DESIGN					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
			CO	KL	M
		UNIT-I			
1.	a).	How Data-Mapping Data onto Aesthetics	1	2	7
	b).	How Cartesian coordinates, and non linear axes used for Data Visualization	1	2	7
		OR			
2.	a).	Explain how color can be used to represent data values and tool to highlight	1	2	7
	b).	Explain Visualizing amounts	1	2	7
		UNIT-II			
3.	a).	How to visualize multiple distributions at the same time	2	3	7
	b).	Explain empirical and highly skewed distributions	2	2	7
		OR			
4.	a).	Explain how visualizing distributions in vertical and horizontal axis	2	2	7
	b).	Explain visualization of distributions by Histograms and Density-plots	2	2	7
		UNIT-III			
5.	a).	Explain the case where side-by-side Bars are suitable than pie charts	3	3	7
	b).	Explain Visualization of Nested Proportions- Mosaic Plots and Treemaps	3	2	7
		OR			
6.	a).	Explain Nested Pies and parallel sets	3	2	7
	b).	Explain Visualization of Associations Among Two or More Quantitative Variables	3	2	7
		UNIT-IV			
7.	a).	Explain how to framing probabilities as frequencies	4	3	7
	b).	Explain Visualizing the Uncertainty of Curve Fits	4	2	7
		OR			

8.	a).	Define Detrending? Explain Time Series Decomposition	4	3	7
	b).	Explain how to visualize response curves	4	2	7
		UNIT-V			
9.	a).	What is principle of proportional ink? Explain Visualization along Linear Axes?	5	2	7
	b).	Explain visualization along Logarithmic Axes	5	2	7
		OR			
10.	a).	Explain Partial Transparency and Jittering	5	2	7
	b).	How to Use 2D Histograms, Contour Lines in visualization	5	2	7

CO-COURSE OUTCOME

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M-MARKS

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